Vanguard Motors — Connected Vehicle Telematics Platform

Case Study. Client: Vanguard Motors AG. Delivered 2024.

The Problem

Vanguard Motors ships 340,000 vehicles a year across fourteen European markets. Every vehicle since the 2019 model year carries a telematics control unit, but the data landed in three separate regional systems that could not be queried together. Warranty engineers investigating a suspected fault were waiting eleven days for a data extract, by which point the affected build batch had already shipped.

What We Built

A single connected-vehicle data platform ingesting telemetry from every vehicle in the fleet, replacing the three regional systems.

The ingestion tier accepts CAN bus summaries, diagnostic trouble codes and GPS position from 2.1 million active vehicles, normalising three generations of telematics hardware into one event schema. Peak sustained throughput is 240,000 events per second.

The warranty analytics workspace lets engineers query the fleet by build batch, component supplier, market and date range, and returns a result in under four seconds against the full history.

An over-the-air update service delivers firmware to targeted vehicle cohorts with staged rollout and automatic rollback on failure-rate threshold breach.

Technology Stack

The ingestion and stream-processing services are written in Rust, chosen for predictable memory behaviour at sustained high throughput. Apache Kafka is the event backbone. ClickHouse stores the time-series telemetry and PostgreSQL holds vehicle and build metadata. The warranty workspace front end is built in React with TypeScript, and the companion mobile application for field engineers is React Native. Everything runs on Microsoft Azure using Kubernetes, with infrastructure defined in Terraform.

Engagement Model

Vanguard engaged us on a dedicated team basis rather than a fixed bid. The regulatory surface across fourteen markets was not fully mapped at kickoff, and a fixed bid would have forced us to price uncertainty we could not yet quantify.

The engagement ran for 16 months. Total contract value was USD 1,240,000, invoiced monthly.

Commercials

Phase	Duration	Invoiced
Discovery and regulatory mapping	Months 1-3	USD 210,000
Ingestion platform	Months 3-9	USD 465,000
Warranty analytics workspace	Months 8-14	USD 390,000
OTA update service and handover	Months 13-16	USD 175,000

Team Structure

Fourteen people at peak: one delivery lead, one solution architect, six Rust engineers, two data engineers, two front-end engineers, one QA automation engineer and one automotive compliance specialist provided by Vanguard.

Security and Compliance

The platform operates under UNECE WP.29 R155 cybersecurity management requirements, which apply to every vehicle type approved in the European Union. All telemetry is pseudonymised at ingest; re-identification requires a dual-authorisation workflow that is itself audited. GDPR data subject access requests are served from a dedicated index with a 72-hour SLA.

Results

Warranty investigation turnaround fell from eleven days to under four hours, measured across 1,900 investigations in the year following rollout.

Early fault detection on two build batches avoided an estimated EUR 4,100,000 in warranty liability and one prospective recall.

Over-the-air updates now reach 94% of targeted vehicles within seven days, against a previous dealer-visit rate of 31% within ninety days.